

CollabMatrix: NASA-Wide Resource, Document & Review Hub

Abstract

The National Aeronautics and Space Administration (NASA) manages complex, multi-year space flight programs using NPR 7120.5, a comprehensive policy framework that defines mandatory lifecycle phases, Key Decision Points (KDPs), and compliance requirements for all space flight programs and projects. This framework governs how projects progress from Pre-Phase A concept studies through Phase F closeout, with formal reviews required at each transition. However, while NPR 7120.5 establishes what must be accomplished, it provides no integrated software platform for tracking compliance, documents, or resources across NASA's geographically distributed centers.

This report presents CollabMatrix, a proposed NASA-wide collaboration hub designed to address the fragmentation in NASA's current project management toolset. CollabMatrix integrates workforce dashboards, document repositories with version control, milestone tracking aligned with NPR 7120.5 phases, and cross-center visibility features. The platform connects the currently disconnected tools—POLARIS (reference database), Comments Export Tool (single-document review tracking), Action Item System (generic task management), and PCEC (Excel-based cost estimation)—into a unified, lifecycle-aware system.

The project scope includes development of a secure, centralized web-based platform with role-based access control, pilot deployment at three NASA centers (Ames, JPL, and Johnson), and integration with existing NASA schedule and cost management tools. Expected outcomes include a 25-30% reduction in document review time, 20% improvement in cross-center project visibility, and streamlined compliance tracking against NPR 7120.5 requirements. This report details the project charter, work breakdown structure, scheduling, resource planning, and lessons learned throughout the project management lifecycle.

Part 1: Background

1.1 About NASA

The National Aeronautics and Space Administration (NASA) is America's civil space program and the global leader in space exploration. Established on October 1, 1958, when President Dwight D. Eisenhower signed the National Aeronautics and Space Act, NASA's primary mission is exploring the secrets of the universe for the benefit of all—investigating the unknown in air and space, innovating for the benefit of humanity, and inspiring the world through discovery (NASA, 2024). The agency comprises a workforce of approximately 18,000 civil servants distributed across 17 centers and facilities, with an additional 60,000 contractors supporting its mission (NASA Organization, 2024).

1.2 NASA Organizational Structure

NASA operates with a hierarchical organizational structure led by the Office of the Administrator. The Administrator, appointed by the President of the United States, provides overall leadership and direction for the agency. Supporting the Administrator are a Deputy Administrator and various Associate Administrators, each responsible for specific Mission Directorates or functional areas (NASA Strategic Management Handbook, 2024). The agency is organized into six Mission Directorates: Aeronautics Research, Exploration Systems Development, Science, Space Operations, Space Technology, and Mission Support (NASA Directorates, 2024).

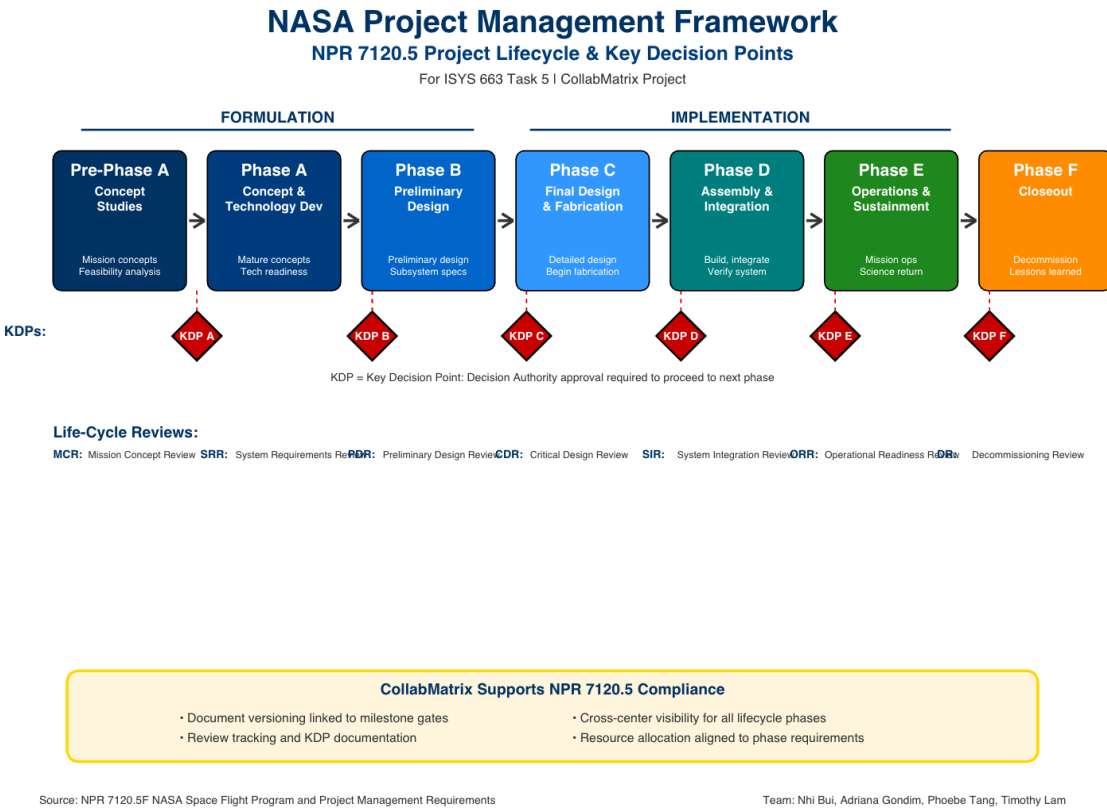
Projects are implemented through NASA's geographically distributed centers, each with specialized capabilities. The Jet Propulsion Laboratory (JPL), managed by Caltech, leads robotic planetary missions, including Mars exploration. Goddard Space Flight Center (GSFC) manages Earth science missions and space telescopes such as the James Webb Space Telescope. Johnson Space Center (JSC) oversees human spaceflight operations and the International Space Station, while Kennedy Space Center (KSC) manages launch operations (NASA Centers and Facilities, 2024).

1.3 NASA Project Management Framework: NPR 7120.5

NASA governs space flight programs and projects through NPR 7120.5, the NASA Space Flight Program and Project Management Requirements. This procedural requirement document defines the lifecycle phases, Key Decision Points (KDPs), roles, responsibilities, and compliance requirements for all NASA space flight activities (NPR 7120.5F,

2024). The framework establishes a systematic approach to managing projects from initial concept through closeout, ensuring technical rigor, cost control, and schedule management across all mission types.

Project Lifecycle Phases



The NASA project lifecycle consists of two major phases—Formulation and Implementation—subdivided into seven specific phases (SEH 3.0, NASA, 2024). Pre-Phase A (Concept Studies) identifies mission concepts and conducts feasibility studies. Phase A (Concept and Technology Development) matures concepts and assesses technology readiness. Phase B (Preliminary Design and Technology Completion) develops preliminary designs and subsystem specifications. Phase C (Final Design and Fabrication) completes detailed design and begins fabrication. Phase D (System Assembly, Integration, and Test) builds, integrates, and verifies the complete system. Phase E (Operations and Sustainment) conducts mission operations and science return. Phase F (Closeout) decommissions the system, captures lessons learned, and archives documentation.

Key Decision Points and Reviews

Progression between lifecycle phases is governed by Key Decision Points (KDPs), formal gates where the Decision Authority evaluates the project's readiness to proceed. At each KDP, management examines technical maturity, resource adequacy, risk posture, and stakeholder alignment (NPR 7123.1D, 2024). Major lifecycle reviews include the System Requirements Review (SRR), Preliminary Design Review (PDR), Critical Design Review (CDR), Operational Readiness Review (ORR), and Flight Readiness Review (FRR). At KDP C, the Agency Baseline Commitment (ABC) is established, forming the baseline against which cost and schedule performance is measured.

Project Team Roles

NPR 7120.5 defines specific roles and responsibilities for project execution (NPR 7120.5F Chapter 3, 2024). The Project Manager holds overall responsibility for cost, schedule, and technical performance, reporting to the Mission Directorate Associate Administrator. The Project Systems Engineer manages technical integration, requirements flow-down, and verification activities. The Program/Project Chief Engineer (PCE) serves as the Engineering Technical Authority, establishing design processes, specifications, and standards. The Project Scientist defines science requirements and leads the science team. Instrument Managers oversee development of specific subsystems, while Safety and Mission Assurance personnel manage risk and compliance verification.

1.4 Example Mission Team Structures

James Webb Space Telescope (JWST)

The James Webb Space Telescope exemplifies NASA's complex, multi-center project structure. JWST is an international collaboration among NASA, the European Space Agency (ESA), and the Canadian Space Agency (CSA). NASA's Goddard Space Flight Center managed overall telescope development, while JPL managed the Mid-Infrared Instrument (MIRI). The primary contractor, Northrop Grumman, was responsible for the spacecraft element and sunshield (JWST Team, NASA, 2024). At peak development, the project involved over 1,000 personnel across 29 U.S. states and 14 countries. Project Manager Bill Ochs managed 400-500 direct team members, emphasizing communication as the critical success factor (APM Interview, 2024).

Mars 2020 Perseverance Rover

The Mars 2020 Perseverance mission demonstrates JPL's approach to planetary exploration projects. NASA's Jet Propulsion Laboratory built and manages rover operations for the Science Mission Directorate. John McNamee served as Project Manager, with Matt Wallace and Jennifer Trosper as Deputy Project Managers. Ken Farley of Caltech served as Project Scientist, supported by Deputy Project Scientists Katie Stack Morgan and Ken Williford (Mars 2020 Press Kit, 2021). The project structure included specialized teams for entry, descent, and landing (led by Al Chen), instrument development, and surface operations.

1.5 The Problem: Fragmented Project Management Tools

While NPR 7120.5 provides a comprehensive policy framework, it is fundamentally a document—not a software platform. NASA has developed individual tools to address specific project management needs, but these tools operate in isolation without integration:

POLARIS (Program/Project On-line Library and Resource Information System) provides searchable access to NPR 7120.5 requirements and interactive lifecycle charts, but functions only as a read-only reference database on NASA's internal intranet (IEEE, 2016).

Comments Export/Management Tool (MFS-33498-1) exports Word document comments to Excel spreadsheets for tracking review disposition, but is limited to single-document scope with no connection to lifecycle phases or cross-center visibility (NASA Software Catalog, 2024).

Action Item System (GSC-16768-1) provides web-based task assignment with email notifications and deadline tracking, but offers generic functionality without NASA-specific lifecycle structure or document linkage (NASA Software Catalog, 2024).

Project Cost Estimating Capability (PCEC) (MFS-33187-2) enables cost estimation using NASA's Cost Estimating Relationships (CERs), but operates as an isolated Excel-based tool without integration to schedule or resource systems (NASA Software Catalog, 2024).

The result is that no single platform connects the critical question chain: "What phase are we in?" → "What documents are required?" → "Who is working on them?" → "Are we ready for the next KDP?" Project managers must manually coordinate between disconnected systems, creating inefficiencies, communication gaps, and compliance tracking challenges across NASA's multi-center, multi-contractor environment.

1.6 Options and Recommendations

Mission Insight X: An integrated analytics platform combining cost models, schedule baselines, EVM data, and risk information into a unified decision-support environment. It enables rapid “what-if” simulations and provides predictive insights to improve cost and schedule management and early trade-space evaluation.

CollabMatrix: A NASA-wide collaboration hub designed to streamline resource management, document control, and review workflows across multiple centers and contractors. It improves visibility, reduces duplication, and aligns documentation and tasks with NPR 7120.5 lifecycle milestones.

KnowledgeGraph 360: An enterprise knowledge-graph system capturing mission heritage, requirements, subsystem relationships, verification results, lessons learned, and personnel expertise. It enhances traceability, supports informed design decisions, and promotes institutional knowledge reuse.

Adaptive Lifecycle Manager: A hybrid project-management platform that merges Agile team workflows with NASA’s phased lifecycle requirements. It maintains end-to-end traceability, provides real-time cost/schedule/risk monitoring, and enables iterative development while remaining compliant with NPR 7120.5.

(AI assistance (ChatGPT) was used to refine wording in the Options & Recommendations)

1.7 The Solution: CollabMatrix

CollabMatrix addresses this integration gap by providing a lifecycle-aware collaboration hub that connects resources, documents, and reviews in a unified platform aligned with NPR 7120.5 requirements. The system maps directly to NASA's lifecycle phases, automatically surfacing required documents and tracking readiness for each KDP. It integrates document review workflows similar to the Comments Export Tool but connected to lifecycle milestones. Action items are tracked in context, linked to specific KDPs and deliverables. Cross-center visibility enables personnel at JPL, GSFC, JSC, and other centers to see unified project status, resource allocation, and document workflows.

Part 2: CollabMatrix: NASA Cross-Center Governance & Collaboration Systems

2.1. Overview of the Project

CollabMatrix revolutionizes NASA's approach to distributed project management by enabling cryptographically-secured, direct team-to-team collaboration while preserving Technical Authority independence mandated by NPR 7120.5. The platform transforms project management culture from coordination burden to collaboration celebration, protects PMs from tactical overhead, and facilitates breakthrough interdisciplinary innovation through AI-enhanced knowledge bridging. Using asymmetric encryption and distributed ledger technology, CollabMatrix creates immutable audit trails that ensure compliance while enabling unprecedented transparency and accountability across NASA's complex multi-center operations.

2.2. How This Connects to CollabMatrix

Based on official NASA sources, GAO reports, and APPEL Knowledge Services documentation, here are the key problems:

1. Information Silos Between Centers

NASA Chief Knowledge Architect David Meza says the biggest challenge is accessing the information due to silos between groups, departments, programs and products. [APPEL Knowledge Services](#) This is a critical finding—it comes from NASA's own knowledge leadership.

2. Cross-Center Collaboration Barriers

In the mid-2000s, the Agency faced major knowledge sharing challenges due to geographically isolated field centers that inhibited engineers from sharing their experiences, expertise, ideas, and lessons learned. [ScienceDirect](#)

Cultural differences between centers, when not presented as center rivalry, most often showed up as differences in processes. These differences led to frustration and confusion, and in some cases, mistrust and an unwillingness to communicate. [APPEL Knowledge Services](#)

3. Communication Breakdowns Lead to Mission Failures

I learned that the root cause of most mishaps, be they in hardware, software, or requirements, could be traced back to miscommunication or a lack of communication. [APPEL Knowledge Services](#)

One manager from the survey relayed a story about a long-running technical disagreement between centers that persisted for months. Finally the teams were brought together, closed into a conference room, and told to solve the problem. They did, less than a half-hour later. [APPEL Knowledge Services](#)

4. Lessons Learned Not Applied Across Projects

The failure of the Mars Polar Lander and Climate Orbiter spacecraft raised concerns that lessons learned from past mishaps and programs were not being effectively applied towards future mission success. [GovInfo](#)

One manager stated that "Until we can adopt a culture that admits frankly to what really worked and didn't work, I find many of these tools to be suspect." Another critical barrier highlighted by the survey is the lack of time available for lessons learned. [GovInfo](#)

5. Workforce Resource Visibility Issues

The GAO cited several cases where projects have run into delays because of "experienced workforce shortages" and difficulty retaining key workers. NASA's workforce, the report noted, is aging, with 56 percent of its employees age 50 or older. [SpaceNews](#)

6. Cost and Schedule Overruns

Since 2023, NASA's cumulative cost and schedule performance has improved. Cost overruns decreased from \$7.6 billion in 2023 to \$4.4 billion in 2024. Schedule overruns decreased from a total of 20.9 years in 2023 to 14.5 years in 2024. [U.S. GAO](#)

While improving, these numbers are still staggering—and much of this stems from project management coordination issues.

7. Requirements Communication Problems

Despite our best intentions, time is always short in the project formulation phase. "Requirements creep" is a common syndrome, and, generally speaking, it is the enemy of both project management and systems engineering. [APPEL Knowledge Services](#)

NASA Problem	CollabMatrix Solution
Information silos between centers	Cross-center visibility dashboard
Cultural/process differences	Standardized workflow aligned to NPR 7120.5
Communication breakdowns	Real-time document and review tracking
Lessons not applied	Integrated milestone and document history
Workforce visibility gaps	Resource tracking showing who is assigned where, workload, and skills
Requirements creep	Version control and audit logs

2.3. Project Timeline Summary:

The CollabMatrix project follows an 18-month implementation schedule from initiation through full deployment across three NASA pilot centers.

Timeline Element	Details
Project Start Date	January 6, 2025
Pilot Launch (3 Centers)	September 30, 2025 (Month 9)
Full Deployment Complete	June 30, 2026 (Month 18)
SA&A Certification	September 30, 2026 (Month 21)
Total Project Duration	21 months (including 3-month operational validation)
Pilot Centers	Ames Research Center, Johnson Space Center, Kennedy Space Center

2. Schedule Analysis

The CollabMatrix project schedule spans approximately 18 months from initiation to full deployment, with an additional 3-month operational validation period for SA&A certification. The **critical path** runs through the following sequence:

Requirements Gathering (1.1.1) → System Architecture Design (1.2.1) → Security Design (1.2.3) → Core Platform Development (1.3.1) → Security Integration (1.3.3) → Internal Testing (1.4.1) → Pilot Deployment (1.4.2) → Final Approval (1.4.5).

This path totals approximately 36 weeks with zero float, meaning any delay in these activities directly impacts the pilot launch date. The PKI/Certificate Authority setup represents the highest-risk activity on the critical path due to NASA cross-certification requirements and potential coordination delays with NASA IT Security. Key milestones include: Design Review Complete (Week 12), Prototype Demo (Week 24), Pilot Launch (Week 36), and Full Deployment (Week 72). Non-critical activities such as UI/UX design and training material development have 2-4 weeks of float and can absorb minor delays without affecting the overall schedule.

3. Risk Register

The following risk register identifies key project risks, their probability and impact, and planned response strategies.

ID	Risk Description	Prob.	Impact	Response Strategy	Owner
R1	PIV smart card integration delays due to NASA IT coordination	High	High	Mitigate: Early engagement with NASA PKI team; parallel development tracks	PKI Architect
R2	User adoption resistance at pilot centers	Medium	High	Mitigate: Change management plan; early stakeholder	Project Manager

				involvement; champion program	
R3	Security Assessment & Authorization (SA&A) delays	High	High	Mitigate: Engage NASA OCIO early; use pre-approved security patterns; iterative compliance reviews	Compliance Officer
R4	Scope creep from stakeholder feature requests	Medium	Medium	Avoid: Strict change control process; formal Change Control Board; documented scope baseline	Project Manager
R5	Integration failures with existing NASA IMS tools	Medium	High	Mitigate: Early API prototyping; dedicated integration testing phase; fallback manual workflows	Systems Engineer
R6	Key personnel turnover during development	Low	High	Transfer: Cross-training program; comprehensive documentation; knowledge transfer sessions	Project Manager

Risk Rating Key: High-High risks (R1, R3) require active monitoring and weekly status updates. Medium-High risks (R2, R5) are reviewed bi-weekly. Yellow highlighting indicates critical path risks.

4. RACI Matrix

The RACI matrix defines roles and responsibilities for key project deliverables: **R** = Responsible (does the work), **A** = Accountable (owns the outcome), **C** = Consulted (provides input), **I** = Informed (kept updated).

Deliverable	PM	Sys Arch	PKI Arch	UI/UX	QA	NASA IT
System Architecture Document	A	R	C	I	C	C
PKI/CA Implementation	A	C	R	I	C	C
UI Dashboard Design	A	C	I	R	C	I
Core Platform Development	A	R	C	C	C	I
Security Integration	A	R	R	I	C	C

NPR 7120.5F Compliance Matrix	A	C	C	I	R	C
System Security Plan (SSP)	A	C	R	I	C	R
User Training Materials	A	I	I	R	C	I
Pilot Deployment	A	R	R	C	R	C
SA&A Certification	A	C	R	I	R	R

Key: PM = Project Manager | Sys Arch = Systems Architect | PKI Arch = PKI Architect | UI/UX = UI/UX Designer | QA = Quality Assurance | NASA IT = NASA Center IT Teams

5. Stakeholder Register

The following table identifies key stakeholders, their roles, interests, and engagement strategies.

Stakeholder	Role	Interest/Influence	Engagement Strategy
NASA Senior Management	Executive Sponsor	High influence; funding approval; strategic alignment	Monthly briefings; KDP gate reviews
Center Project Managers	Primary End Users	High interest; daily system usage; efficiency gains	Weekly demos; feedback sessions; champion program
Center Directors (Ames, JSC, KSC)	Pilot Site Leads	High influence; resource allocation; center adoption	Quarterly reviews; pilot success metrics
NASA OCIO / IT Security	Technical Authority	High influence; SA&A approval; security compliance	Bi-weekly security reviews; early architecture input
NASA OIG	Compliance Oversight	Medium influence; audit trail requirements	API access for audit; compliance documentation
Engineers & Scientists	End Users	High interest; document management; collaboration	User acceptance testing; training sessions
Contractors	External Collaborators	Medium interest; cross-team visibility	Restricted access training; onboarding guides
Development Team	Implementation	High interest; technical decisions; workload	Daily standups; sprint planning; retrospectives

6. CollabMatrix Project Scope

The CollabMatrix project will design, develop, test, and deploy a highly secure, web-based collaboration platform that supports NASA's complex, multi-center project management environment. The system will leverage public key infrastructure (PKI) and asymmetric encryption to enforce strict compliance with NPR 7120.5F requirements while enabling seamless task verification and cross-team collaboration. The platform will include the following key features:

- **In Scope:**
 - **Certificate Authority (CA):** CollabMatrix will establish a dedicated CA that issues and manages digital certificates for all users in accordance with NASA's PKI policy. The CA will be cross-certified with the NASA Root CA to ensure agency-wide trust.
 - **Digital Signature-Based Task Verification:** All task completions and approvals in CollabMatrix will require digital signatures using NASA PIV smart cards. The system will maintain an immutable record of all signatures, providing non-repudiation and ensuring accountability.
 - **Multi-Signature Approval Schemes:** CollabMatrix will enforce multi-signature approval for critical tasks, ensuring that all required authorities (e.g., Project Manager, Chief Engineer) have digitally signed off before a task is considered approved.
 - **Cross-Team Collaboration:** The platform will enable secure cross-team collaboration through digitally signed requests and approvals. CollabMatrix will create secure collaboration spaces for teams to work together on shared tasks while maintaining strict access controls.
 - **Work Breakdown Structure (WBS) Integration:** CollabMatrix will automatically update the project's WBS as tasks are completed and approved, displaying the digital signatures associated with each task for transparency and auditability.
 - **Comprehensive Audit Trail:** The system will maintain a tamper-proof audit trail of all task actions, approvals, and collaborations, including digital signatures, timestamps, and role attributes. Secure APIs will be provided for NASA OIG to access the audit trail for compliance monitoring.

- **NPR 7120.5F Compliance Mapping:** CollabMatrix will include a compliance mapping module that links each platform feature to specific NPR 7120.5F requirements, enabling project teams to demonstrate compliance through system usage.

- **Out of Scope:**

- **External PKI Integration:** While CollabMatrix will cross-certify with the NASA Root CA, it will not integrate with external PKI systems from other agencies or partners. Collaborations with external entities will be handled through separate access control mechanisms.
- **Hardware Token Support:** CollabMatrix will rely on NASA's existing PIV smart card infrastructure for user authentication and digital signatures. The platform will not support additional hardware tokens or biometric devices.
- **Classified Data Handling:** The initial release of CollabMatrix will be designed for use with sensitive but unclassified (SBU) data only. The system will not be certified for handling classified information. Separate systems will be used for any classified project activities.

The project will be considered complete when CollabMatrix has been successfully deployed to the pilot centers, meets all the PKI-related requirements within the defined scope, and has passed a rigorous security assessment and authorization (SA&A) process. The system will be considered operational after 3 months of successful use with no Critical/High security or compliance defects.

The project team will include a dedicated PKI Architect who will design the CollabMatrix CA and the digital signature workflows in alignment with NASA policies. The team will also have a Compliance Officer who will map NPR 7120.5F requirements to system features and coordinate with the OIG for audit readiness.

The use of PKI and digital signatures sets CollabMatrix apart from traditional collaboration tools and demonstrates a deep commitment to meeting NASA's unique security and regulatory requirements. It shows that the system is designed from the ground up to enforce NPR 7120.5F compliance, rather than treating it as an afterthought.

As we engage with stakeholders across NASA, we can highlight these PKI capabilities as a key value driver. The ability to provide non-repudiation, tamper-proof audit trails, and cryptographically enforced segregation of duties will be highly attractive to project managers, center directors, and compliance auditors alike.

7. Work Breakdown Structure (WBS)

The following WBS decomposes the CollabMatrix project into manageable work packages aligned with the project lifecycle.

1.0 CollabMatrix: NASA-Wide Resource, Document & Review Hub

1.1 Initiation & Planning

- 1.1.1 Stakeholder Requirement Gathering – Identify pilot center stakeholders and collect system needs
- 1.1.2 Define Goals & Objectives – Establish project purpose, success criteria, and measurable outcomes
- 1.1.3 Data Collection & Baseline Assessment – Audit existing tools, document workflows, and data repositories
- 1.1.4 Governance & Compliance Planning – Map NASA IT policies, define security and data handling standards
- 1.1.5 Project Charter & Approval – Formalize project charter, roles, and initial budget

1.2 Design & Architecture

- 1.2.1 System Architecture Design – Develop system blueprint (secure web architecture, centralized database, APIs)
- 1.2.2 UI/UX & Dashboard Design – Create dashboard mockups for resource tracking, document management, and milestones
- 1.2.3 Security Design – Implement authentication, encryption, and compliance measures
- 1.2.4 Role-Based Access Control (RBAC) Framework – Define access levels for NASA staff, contractors, and admins
- 1.2.5 Review & Approval of Design Documents – Internal review by project managers and NASA IT leads

1.3 Implementation & Development

- 1.3.1 Core Platform Development – Build collaboration hub backend and web-based interface
- 1.3.2 Dashboard & Resource Modules – Develop document management, milestone tracking, and reporting modules
- 1.3.3 Security & Access Control Integration – Implement RBAC, password protection, and logging features
- 1.3.4 Prototype Creation & Internal Demo – Create functional prototype for internal NASA team validation

1.4 Testing & Quality Assurance

- 1.4.1 Internal Testing – Conduct functionality, performance, and security testing
- 1.4.2 Pilot Deployment (Ames, JSC, KSC) – Deploy the system for pilot centers
- 1.4.3 User Feedback Collection – Gather feedback from pilot users and project managers

- 1.4.4 Bug Fixes & Iterative Improvements – Address issues and optimize based on pilot testing
- 1.4.5 Final Approval for Limited Launch – Obtain management sign-off for pilot phase completion

1.5 Training & Support

- 1.5.1 Conduct Pilot User Training Sessions – Deliver training to NASA center users and admins
- 1.5.2 Provide User Manuals & Documentation – Finalize and distribute support materials
- 1.5.3 Post-Deployment Support – Monitor system performance, troubleshoot, and refine workflows

1.6 Project Closeout

- 1.6.1 Final Evaluation & Lessons Learned – Review pilot success metrics and capture lessons
- 1.6.2 Documentation Handover – Deliver technical and user documentation to NASA IT
- 1.6.3 Project Closure & Handoff – Archive project files and prepare for potential NASA-wide phase

Detail:

https://app.diagrams.net/#G1ini7lC3TSBTkuqXIZo1pO1amYRDT_xC9#%7B%22pageId%22%3A%22ZPn40aEiryVpxYKQIgFf%22%7D

8. Gantt Chart

WBS	Task Name	Duration	Start	Finish	Predecessors	Successors	Resource Names
0	Project: CollabMatrix	161 days	Wed 11/5	Tue 6/23/26			
1	CollabMatrix	161 days	Wed 11/5	Tue 6/23/26			
1.1	Initiation	9 days	Wed 11/5	Mon 11/17/25			
1.1.1	Stakeholder Identification	3 days	Wed 11/5	Fri 11/7/25		4	Systems Lead
1.1.2	Project Charter Development	5 days	Mon 11/10	Fri 11/14/25	3	5	Operations Lead
1.1.3	Kick-off Meeting	1 day	Mon 11/17	Mon 11/17/25	4	9,10,8	Systems Lead,Operations Lead
1.2	Planning	24 days	Tue 11/18	Fri 12/19/25			
1.2.1	Requirement Gathering	8 days	Tue 11/18	Thu 11/27/25		16,17,18,19,20,12,13,14,21	
1.2.1.1	Functional Requirements	8 days	Tue 11/18	Thu 11/27/25	5		Product Lead
1.2.1.2	Security Requirements	8 days	Tue 11/18	Thu 11/27/25	5		Systems Lead
1.2.1.3	Compliance Requirement	8 days	Tue 11/18	Thu 11/27/25	5		Operations Lead
1.2.2	System Design	10 days	Fri 11/28	Tue 12/11/25		19,20,21	Systems Lead
1.2.2.1	Architecture Design	10 days	Fri 11/28	Tue 12/11/25	7		Systems Architect
1.2.2.2	Database Design	8 days	Fri 11/28	Tue 12/9/25	7		Systems Database Engineer
1.2.2.3	UI/UX Design	10 days	Fri 11/28	Tue 12/11/25	7		Product UX/UI Researcher
1.2.3	PKI Design	6 days	Fri 11/28	Fri 12/5/25		19,20,21	Product Lead
1.2.3.1	Certificate Authority	6 days	Fri 11/28	Fri 12/5/25	7		Product CA Engineer
1.2.3.2	Digital Signature Workflow Design	5 days	Fri 11/28	Tue 12/4/25	7		Product Architect
1.2.3.3	Smart Card Intergration Design	5 days	Fri 11/28	Tue 12/4/25	7		Product Engineer
1.2.4	Project Schedule Development	5 days	Fri 12/12	Tue 12/18/25	7,11,15	21	Product Lead,Systems Lead
1.2.5	Resource Planning	5 days	Fri 12/12	Tue 12/18/25	7,11,15	21	Operations Lead
1.2.6	Review/Approval Center - Level Stakeholders	1 day	Fri 12/19	Fri 12/19/25	19,20,7,11,15	24,25,26,29,33,34,35	Product Lead,Operations Lead
1.3	Execution	85 days	Mon 12/22	Wed 4/22/26			
1.3.1	CollabMatrix Software Development	45 days	Mon 12/22	Wed 2/25/26		37	Systems Lead
1.3.1.1	Backend Development	30 days	Mon 12/22	Wed 2/4/26	21	27	Backend Developer
1.3.1.2	Frontend Development	30 days	Mon 12/22	Wed 2/4/26	21	27	Frontend Developer
1.3.1.3	Database Development	20 days	Mon 12/22	Wed 1/21/26	21	27	Systems Database Engineer
1.3.1.4	Intergration	15 days	Thu 2/5/26	Wed 2/25/26	24,25,26		Systems Architect
1.3.2	PKI Implementation	25 days	Mon 12/22	Wed 1/28/26		37	Product Lead
1.3.2.1	CA Setup and Configuration	10 days	Mon 12/22	Wed 1/7/26	21	30,31	Product CA Engineer
1.3.2.2	Digital Signature Intergration	15 days	Thu 1/8/26	Wed 1/28/26	29		Product Architect
1.3.2.3	Smart Card Intergration	15 days	Thu 1/8/26	Wed 1/28/26	29		Product Engineer
1.3.3	Compliance Feature Development	15 days	Mon 12/22	Wed 1/14/26		37	Operations Lead
1.3.3.1	NPR 7120.5F Requirement Mapping	15 days	Mon 12/22	Wed 1/14/26	21		Operations Compliance
1.3.3.2	Audit Trail Implementation	15 days	Mon 12/22	Wed 1/14/26	21		Operations Administrator
1.3.3.3	Access Control Implementation	15 days	Mon 12/22	Wed 1/14/26	21		Operations Security
1.3.4	Prototype Creation & Internal Demo	5 days	Thu 2/26	Wed 3/4/26		41,42,43,50,51,52,53	
1.3.4.1	Prototype Development	3 days	Thu 2/26	Mon 3/2/26	23,28,32	38,39	Systems Architect,Backend
1.3.4.2	Review/Approval - Internal	1 day	Tue 3/3/26	Tue 3/3/26	37	39	Operations Lead,Product Lead
1.3.4.3	Review/Approval - Stakeholder User	1 day	Wed 3/4/26	Wed 3/4/26	37,38		Operations Lead,Product Lead
1.3.5	Testing	22 days	Thu 3/5/26	Fri 4/3/26		55,50,51,52,53,46,47	Product Lead
1.3.5.1	Unit Testing	12 days	Thu 3/5/26	Fri 3/20/26	36	44	Systems Architect,Systems Lead
1.3.5.2	Intergration Testing	12 days	Thu 3/5/26	Fri 3/20/26	36	44	Product Architect,Product Engineer
1.3.5.3	Security Testing	12 days	Thu 3/5/26	Fri 3/20/26	36	44	Operations Security,Systems Architect
1.3.5.4	User Acceptance Testing	10 days	Mon 3/23	Fri 4/3/26	41,42,43		Backend Developer,Frontend Developer
1.3.6	Test Assessment	13 days	Mon 4/6/26	Wed 4/22/26			
1.3.6.1	Stakeholder Feedback	3 days	Mon 4/6/26	Wed 4/8/26	40	48	Product Lead,Operations Lead
1.3.6.2	Technical Review	1 day	Mon 4/6/26	Mon 4/6/26	40	48	Systems Lead
1.3.6.3	Product Evaluation & Improvement	2 wks	Thu 4/9/26	Wed 4/22/26	46,47		Product Lead,Operations Lead
1.3.7	Documentation	10 days	Mon 4/6/26	Fri 4/17/26		55	Systems Lead
1.3.7.1	User Guides	10 days	Mon 4/6/26	Fri 4/17/26	36,40		Technical Writer
1.3.7.2	Training Material	10 days	Mon 4/6/26	Fri 4/17/26	36,40		Technical Writer
1.3.7.3	FAQ Troubleshooting	8 days	Mon 4/6/26	Wed 4/15/26	36,40		Technical Writer
1.3.7.4	API Documentation	10 days	Mon 4/6/26	Fri 4/17/26	36,40		Technical Writer
1.4	Deployment	33 days	Mon 4/20	Thu 6/4/26		61	
1.4.1	Pilot Deployment Planning	8 days	Mon 4/20	Wed 4/29/26	40,49	56	Product Architect,Product Engineer
1.4.2	Infrastructure Setup	10 days	Thu 4/30	Wed 5/13/26	55	57	Systems Architect,Systems Lead
1.4.3	CollabMatrix Installation	5 days	Thu 5/14	Wed 5/20/26	56	58,59	Backend Developer,Frontend Developer
1.4.4	Data Migration	10 days	Thu 5/21	Thu 6/4/26	57		Operations Compliance,Operations Lead
1.4.5	User Training	10 days	Thu 5/21	Thu 6/4/26	57		Operations Administrator,Operations Lead

1.5	Closing	13 days	Fri 6/5/26 Tue 6/23/26			
1.5.1	Project Closeout Meeting	1 day	Fri 6/5/26 Fri 6/5/26	54	62,63	Systems Lead, Operations Lead
1.5.2	Lessons Learn Documentation	1 day	Mon 6/8/26 Mon 6/8/26	61	64	Systems Lead, Operations Lead
1.5.3	Final Project Report	5 days	Mon 6/8/26 Fri 6/12/26	61	64	Systems Lead, Operations Lead
1.5.4	Archive Project Document	7 days	Mon 6/15/26 Tue 6/23/26	62,63		Systems Lead, Operations Lead

The detailed project schedule, prepared in Microsoft Project, is included as a separate submission and should be reviewed alongside this report.

9. Primary Objective:

Develop and deploy a secure, centralized collaboration platform that enables NASA centers and contractors to efficiently manage resources, documents, and review processes, aligned with NASA Procedural Requirements (NPR 7120.5). The goal is to improve the efficiency of product maturity across NASA centers.

TABLE 3.0-1 SE Product Maturity from NPR 7123.1

		Formulation				Implementation					
Products	Uncoupled/ Loosely Coupled	KDP 0		KDP I	Periodic KDPs						
	Tightly Coupled Programs	KDP 0			KDP I	KDP II		KDP III		Periodic KDPs	
	Projects and Single Project Programs	Pre- Phase A	Phase A		Phase B	Phase C		Phase D		Phase E	Phase F
		KDP A	KDP B		KDP C	KDP D		KDP E		KDP F	
		MCR	SRR	MDR/SDR	PDR	CDR	SIR	ORR	FRR	DR	DRR
Stakeholder identification and		**Baseline	Update	Update	Update						
Concept definition		**Baseline	Update	Update	Update	Update					
Measure of effectiveness definition		**Approve									
Cost and schedule for technical		Initial	Update	Update		Update	Update	Update	Update	Update	Update
SEMP¹		Preliminary	**Baseline	**Baseline	Update	Update	Update				
Requirements		Preliminary	**Baseline	Update	Update	Update					
Technical Performance Measures definition				**Approve							
Architecture definition				**Baseline							
Allocation of requirements to next lower level				**Baseline							
Required leading indicator trends				**Initial	Update	Update	Update				
Design solution definition				Preliminary	**Preliminary	**Baseline	Update	Update			
Interface definition(s)				Preliminary	Baseline	Update	Update				
Implementation plans (Make/ code, buy, reuse)				Preliminary	Baseline	Update					
Integration plans				Preliminary	Baseline	Update	**Update				
Verification and validation plans		Approach		Preliminary	Baseline	Update	Update				
Verification and validation results						**Initial	**Preliminary	**Baseline			
Transportation criteria and instructions						Initial	Final	Update			
Operations plans					Baseline	Update	Update	**Update			
Operational procedures						Preliminary	Baseline	**Update	Update		
Certification (flight/use)								Preliminary	**Final		
Decommissioning plans					Preliminary	Preliminary	Preliminary	**Baseline	Update	**Update	
Disposal plans					Preliminary	Preliminary	Preliminary	**Baseline	Update	Update	**Update

** Item is a required product for that review

¹ SEMP is baselined at SRR for projects, tightly coupled programs and single-project programs, and at MDR/SDR for uncoupled, and loosely coupled programs.

10. SMART Breakdown:

- **Specific:** Create a digital workspace integrating resource dashboards, document repositories, and milestone tracking.
- **Measurable:** Achieve a 30% reduction in document review time and a 20% improvement in cross-center project visibility within 6 months of pilot launch.
- **Acceptable:** Meets requirements of NASA program managers, engineers, and document control teams.
- **Realistic:** Leverages existing NASA IT infrastructure
- **Time-Based:** Pilot ready within 9 months, full deployment across 3 NASA centers within 18 months.

(AI assistance (ChatGPT), and Professor helped refine wording in the Project Objectives)

11. Project Deliverables:

1. **CollabMatrix Software Application:** A fully functional, web-based collaboration platform that meets all the requirements outlined in the project scope. The application will be developed using modern, secure web technologies and will be hosted on NASA's cloud infrastructure (e.g., NASA Cloud Services).

2. **Public Key Infrastructure (PKI) Components:**

https://nodis3.gsfc.nasa.gov/npg_img/N_PR_2841_0001_/N_PR_2841_0001_.pdf

- **CollabMatrix Certificate Authority (CA):** A dedicated CA that issues and manages digital certificates for all CollabMatrix users in accordance with NASA's PKI policy. The CA will be cross-certified with the NASA Root CA.
- **Digital Signature Workflows:** Secure workflows for task verification, approval, and cross-team collaboration using digital signatures and multi-signature schemes.
- **Smart Card Integration:** Integration with NASA's Personal Identity Verification (PIV) smart cards for user authentication and digital signing.

3. **Compliance Documentation:**

- **NPR 7120.5F Compliance Matrix:** A detailed matrix that maps each CollabMatrix feature to specific NPR 7120.5F requirements, demonstrating how the system enforces compliance.

- **System Security Plan (SSP):** A comprehensive document describing the security controls implemented in CollabMatrix, aligned with NASA's security policies and the NIST SP 800-53 standard.
- **Privacy Impact Assessment (PIA):** An assessment of how CollabMatrix collects, uses, and protects personally identifiable information (PII) in accordance with NASA's privacy policies.
- 4. **User Documentation:**
 - **User Guides:** Step-by-step instructions for using CollabMatrix, tailored to different user roles (e.g., Project Managers, Engineers, Scientists).
 - **Training Materials:** Presentations, videos, and hands-on exercises to support user training and adoption.
 - **FAQ and Troubleshooting Guide:** A knowledge base of common questions and issues, with solutions and workarounds.
- 5. **API Documentation:** Technical documentation for the CollabMatrix APIs, including the audit trail API for OIG access and any integration APIs for connecting with other NASA systems (e.g., financial systems, HR databases).
- 6. **Deployment Packages:** Installation packages and deployment scripts for installing CollabMatrix on NASA's cloud infrastructure, including any necessary configuration files and security certificates.
- 7. **Test Plans and Results:** Detailed test plans and test reports covering unit testing, integration testing, security testing, and user acceptance testing (UAT) of CollabMatrix.
- 8. **Project Management Artifacts:** Project charter, work breakdown structure (WBS), project schedule, risk register, and other relevant project management documents.

These objectives and deliverables collectively ensure that CollabMatrix meets its goal of providing a secure, compliant, and user-friendly collaboration platform for NASA's project teams. The emphasis on PKI, digital signatures, and automated compliance mapping distinguishes CollabMatrix from generic project management tools and positions it as a purpose-built solution for NASA's unique requirements.

Throughout the project lifecycle, the CollabMatrix team will work closely with stakeholders from NASA's Mission Directorates, Center IT departments, and the Office of Inspector General (OIG) to validate the deliverables and ensure alignment with agency policies and expectations.

Primary Framework Documents

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